Jafar Banar

Senior Data & AI Engineer | Python · LLM · ML · 5G/6G | PhD candidate, Telecom & Radar (Chalmers)

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PROFESSIONAL SUMMARY

Senior Data and AI/ML Engineer with 10+ years across telecom, radar/perception, automotive data systems and AI/ML. Senior Data Engineer consultant at Sigma Embedded Engineering Yora AB, most recently on assignment at Volvo Cars on the production test-lab data pipeline (Python, SQL, Azure/AKS): deterministic data synchronization, reversible cleanup and HIL rig onboarding. Previously built and tuned an on-prem LLM/RAG platform on CPU (response time from 5–8 s to about 2.4–3.2 s on short prompts, memory from 12 GB to 8 GB) and found and fixed a radar perception bug at Einride, validated against ground truth. Dual-degree PhD candidate (Chalmers / TU Eindhoven) in synchronization for distributed 5G/6G MIMO and coherent radar. Independent work on agentic AI with verifiable output (groundcite on PyPI).

CORE SKILLS

- **AI/ML & Data Engineering:** LLM/RAG, embeddings and vector search, BM25, agentic systems (LangGraph, MCP), structured-output validation, LLM evaluation, LoRA/QLoRA fine-tuning (A100, SLURM), PyTorch, XGBoost, scikit-learn, ONNX quantization (INT8/FP16), ETL/ELT pipelines, automotive test data, data quality
- **Backend & Cloud:** Python, SQL, FastAPI, Redis, PostgreSQL, Qdrant, FAISS/ChromaDB, Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Linux, Git, Azure, AWS, Grafana
- **Automotive / Radar / Embedded:** Automotive radar, ADAS, sensor fusion, C/C++, Go, ROS, LCM, OpenCV, CAN, ISO 26262, SOTIF, FMCW radar, bistatic radar
- **Telecom / Wireless:** 4G/5G/6G, 3GPP, OFDM, distributed MIMO, beamforming, precoding, SDR/USRP, KPI/QoS analysis, synchronization, calibration
- **Languages & Licence:** English (Fluent), Swedish (Intermediate, SFI C completed, studying SFI D) · Driving licence B

PROFESSIONAL EXPERIENCE

Senior Data Engineer (Consultant) May 2026 – Present
Sigma Embedded Engineering Yora AB, *Gothenburg, Sweden*

- Client assignment at **Volvo Cars** as **Senior Data Engineer** (May to Sep 2026) on the production pipeline that synchronises and delivers measurement data from engine, e-motor, HIL and battery test rigs.
- Made the pairing of two measurement systems deterministic (same sync index first, nearest-timestamp fallback within 24 h, every fallback flagged), turning a silent zero-candidate failure into exactly one reproducible pair; validated on historical rig runs and merged.
- Replaced age-based hard deletes with a reversible, quarantine-first cleanup (per-item logs, rollback, QA dry run before production), bounded hanging report runs with a timeout and stage timings, and onboarded a new dSpace HIL rig with no regression on existing flows.
- Since Sep 2026, building AI demos for Sigma: retrieval-based support assistants, and OpenVINO vision cutting CPU time per image from 124 to 73 ms.

Tech: *Python, SQL, ASAM MDF4, Sympathy for Data, dSpace HIL, Azure/AKS, GitHub Actions, Grafana, Oracle, MongoDB, CIFS/NAS*

Senior Development Engineer (AI Engineer) Feb 2025 – Jul 2026
QRTech AB, *Gothenburg, Sweden*

- Built and tuned an on-prem LLM/RAG platform on a CPU-only server (llama.cpp, 4-bit models, FastAPI, FAISS, Open WebUI): raised threads from 8 to 24, enlarged batches and preloaded the main models, cutting response time from 5–8 s to about 2.4–3.2 s on short prompts and memory from 12 GB to 8 GB.
- Contributed to a team-built LangGraph test-specification generator (7 standards, 3 markets, 5 artifact formats, 8 API endpoints), building its standards-upload API, document-classification UI, documentation and demo.
- Wrote the system requirement specification, interface control document and verification plan for an agricultural robot MVP and contributed to its hazard analysis and risk assessment (HARA).
- Built document anonymization (PII detection with regex, spaCy and Microsoft Presidio, incl. Swedish personal identity numbers, with a gate that blocks external AI calls on non-anonymized text) and an AI CV-matching workflow (embeddings with TF-IDF fallback, n8n).
- Client assignment at **Einride** as **Senior Perception Engineer** (Sep 2025 to Dec 2025), via consultancy. Found and fixed a Go perception bug that under-reported target speed by $\cos^2$(elevation) (3.0 m/s read as 0.75 m/s at 60°), merged after validation on 22 scenarios (~360k radar clusters); proposed a hybrid RANSAC, weighted-SVD and Kalman estimator.
- Einride: replay and KPI workflows against OXTS ground truth, LiDAR timing analysis (1.86M+ packets), detection-limit checks and a SOTIF LiDAR+radar prototype.

Tech: *Python, FastAPI, Redis, LangGraph, FAISS/ChromaDB, Qdrant, PostgreSQL, spaCy, Presidio, n8n, Docker/Kubernetes, GitHub Actions, Jenkins; Einride assignment: Go, C++, Python, ROS1, LCM, PyTorch, OpenCV, OXTS, MCAP/Foxglove, SOTIF*

Founder, CEO & AI Engineer (part-time) Jul 2024 – Present
AIOpt AB, *Gothenburg, Sweden*

- Built a safety and evidence layer for AI-driven changes to 4G/5G networks: a seven-gate deterministic decision engine that approves, escalates or refuses each proposed change, checked against the official 3GPP TS 28.532 OpenAPI by its own tests and run end-to-end in the lab on an open-source 5G software stack (srsRAN-based gNB, Open5GS core), at ~9 ms mean decision latency.
- Designed a tamper-evident evidence format (append-only, hash-chained, Ed25519 countersigned, publicly timestamped) with an open-source offline verifier (MIT, github.com/JafarBanar/aiopt-verify) that shares no code with the writer; KPI forecasting, anomaly detection and root-cause analysis in PyTorch/XGBoost. Almi verification funding (50k SEK), EUIPO SME Fund, Chalmers Innovation Office novelty search, NVIDIA Inception.

Visiting Researcher, Automotive Radar (PhD secondment) Jun 2021 – Nov 2022
NXP Semiconductors, *Eindhoven, Netherlands*

- Seconded from Chalmers (EU MyWave, Marie Skłodowska-Curie) to develop synchronization and signal-processing algorithms for coherent distributed automotive radar used in ADAS.
- Developed a maximum-likelihood estimator for the clock drift between two bistatic 78 GHz FMCW radars, compensated in software and benchmarked against the Cramér-Rao bound (IEEE ISPCS 2024); prepared radar algorithm models for embedded deployment with hardware and software teams.

Tech: *MATLAB, C++, Python, LabVIEW, signal processing, automotive radar, embedded deployment*

Doctoral Researcher & Project Assistant (Wireless Systems) Jul 2020 – Jan 2025
Chalmers University of Technology, *Gothenburg, Sweden*

- Marie Skłodowska-Curie fellow (EU MyWave), dual degree with TU Eindhoven; Project Assistant in Jan 2025.
- Derived closed-form spectral efficiency of distributed massive MIMO under access-point phase noise (EuMC 2022) and developed user-centric phase synchronization with one-index feedback (+85% 5th-percentile spectral efficiency in simulation; journal paper in preparation); built cross-validated MATLAB/Python simulators and ran USRP experiments.
- Supervised 15 thesis students (7 MSc theses, 2 bachelor projects incl. FMCW radar with SDR) with Saab, Ericsson, Qamcom, KTH, Infotiv and others; TPC member for IEEE WCNC 2024–2026 and IEEE ISAC 2026.

Tech: *MATLAB, Python, SDR/USRP, distributed MIMO, OFDM, synchronization, calibration, beamforming, precoding*

RF Optimization Engineer Nov 2019 – Jun 2020
AEC - RavanErtebat (Access Endless Communication)

- Optimized a live Ericsson 2G/3G/4G-LTE network for a large mobile operator: daily KPI reporting, parameter-tuning trials incl. a power-offset trial to cut dropped calls, NodeB audits and trouble-ticket root-cause analysis.

Tech: *Ericsson RAN (RNC/NodeB), TEMS Discovery, Linux, Bash, KPI analysis, RF planning, drive testing*

Telecom Researcher Apr 2016 – Nov 2019
ICT Research Institute

- Researched and simulated 5G and wireless technologies (massive MIMO, full duplex, satellite links, IoT) in MATLAB, Python and C++, checking performance claims numerically rather than repeating them.
- Wrote technical reports and gave internal trainings on 5G, IoT, optical networking, big data and cloud computing.

Tech: *MATLAB, Python, C++, telecom systems analysis, 3GPP/ETSI standards, simulation*

SELECTED PROJECTS

- Published groundcite on PyPI, a zero-dependency Python library and CLI that checks an LLM answer cites only its supplied sources and fails CI on violation; extracted from a citation-grounded 3GPP 5G NR assistant with a from-scratch BM25 index (MRR 0.957, hit-rate@1 0.913 on 23 hand-labelled questions).
- Fine-tuned Qwen2.5-7B with LoRA on an A100 (Chalmers Alvis, SLURM, Apptainer) for telecom root-cause classification: 66.7% vs 40% for rules, while an XGBoost baseline reached 74.9% and was recommended.
- Built AI agents (inbox triage with human approval, invoice extraction, uptime monitoring) with schema-validated output and rule-based fallbacks, and an ONNX edge pipeline (INT8/FP16 quantization, p50/p95 benchmarks, CI latency gate).

EDUCATION

PhD, Electrical Engineering (Dual Degree) Jul 2020 – Feb 2027 (expected)
Chalmers University of Technology / TU Eindhoven, *Sweden / Netherlands*

- Thesis: array calibration and synchronization for distributed massive MIMO and radars; defence February 2027.
- MSc, Electrical Engineering, IUST, 2014–2016: full-duplex cooperative OFDMA networks (journal paper 2017).
- BSc, Electrical Engineering, AUT, 2009–2014: data transmission protocols for LEO satellite links.
- 3rd Place, IEEE ISIT 2025 ML challenge (CSI localization); OxML and MLx 2024 (AI for Global Goals), BSE LLM course.